

3.2.7	Extraction of Metals	
	<i>Principles of metal extraction</i>	know that metals are found in ores, usually as oxides or sulfides and that sulfide ores are usually converted into oxides by roasting in air understand the environmental problems associated with the conversion of sulfides into oxides and also that the sulfur dioxide produced can be used to manufacture sulfuric acid understand that extraction of metals involves reduction understand that carbon and carbon monoxide are cheap and effective reducing agents that are used in the extraction of iron, manganese and copper (reduction equations and conditions only) know why carbon reduction is not used for extraction of titanium, aluminium and tungsten understand how aluminium is manufactured from purified bauxite (energy considerations, electrode equations and conditions only) understand how titanium is extracted from TiO_2 via TiCl_4 (equations and conditions only: either Na or Mg as a reducing agent) understand how tungsten is extracted from WO_3 by reduction with hydrogen (equation, conditions and risks only)
	<i>Environmental aspects of metal extraction</i>	understand the environmental and economic advantages and disadvantages of recycling scrap metals compared with the extraction of metals understand the environmental advantages of using scrap iron to extract copper from aqueous solutions compared with the high-temperature carbon reduction of copper oxide know that the usual source of such aqueous solutions is low grade ore